Security Operations Center (SOC) Monitoring and Log Analysis Report

Avinash Shankar P T
Cyber Security Intern

Abstract

Modern organizations rely on centralized security monitoring systems to detect cyber threats and respond to security incidents efficiently. A **Security Operations Center (SOC)** plays a vital role in monitoring network activity, analyzing logs, and identifying malicious behavior. This report explores the fundamental concepts of SOC operations, security monitoring, and log management.

The experiment demonstrates the implementation of a log collection pipeline using Fluentd to collect system logs and forward them to the Elastic Stack for centralized monitoring and analysis. Security queries were performed using Kibana to detect suspicious events such as failed login attempts (Event ID 4625). Additionally, log normalization was implemented using Logstash to convert Apache access logs into structured JSON format for efficient analysis.

The study also covers key SOC tools, security monitoring techniques, incident response workflows, and documentation practices used by security analysts. The results demonstrate how log analysis and SIEM platforms help organizations detect potential security threats and improve incident response capabilities.

Introduction

Cybersecurity threats are increasing rapidly as organizations rely heavily on digital infrastructure for daily operations. To protect systems and sensitive information from cyberattacks, organizations implement a **Security Operations Center (SOC)** that continuously monitors networks, systems, and applications.

A SOC is responsible for detecting security incidents, analyzing suspicious activities, and responding to potential threats in real time. Security analysts in a SOC environment use various monitoring tools such as Security Information and Event Management (SIEM) platforms to collect and analyze logs generated from different systems.

This report focuses on understanding SOC operations and implementing a basic security monitoring pipeline. The practical component involves collecting system logs using Fluentd and forwarding them to the Elastic Stack for centralized analysis. Security queries were used to detect failed login attempts, which are commonly associated with brute-force attacks.

Additionally, Apache web server logs were processed and normalized into JSON format using Logstash to demonstrate how raw log data can be standardized for better analysis. The objective of this exercise is to gain practical experience with log collection, log processing, and security monitoring techniques commonly used in SOC environments.

Objectives

The main objectives of this experiment are:

1. To understand the fundamentals of **Security Operations Center (SOC) operations**.
2. To study security monitoring concepts and SIEM-based threat detection.
3. To implement a **log collection pipeline** using Fluentd on Ubuntu.
4. To forward system logs to Elastic Stack for centralized monitoring.
5. To practice writing **Kibana Query Language (KQL)** queries to detect failed login events.
6. To perform **log normalization** using Logstash.
7. To understand the workflow of incident detection, investigation, and response in SOC environments.

Log Management Fundamentals

Log management is a critical component of cybersecurity operations because logs provide detailed records of system activities, user actions, and network events. In a **Security Operations Center (SOC)**, log management enables security analysts to monitor system behavior, detect suspicious activities, and investigate security incidents effectively. By collecting and analyzing logs from multiple sources, organizations can identify unauthorized access attempts, system errors, and potential cyberattacks.

Log Lifecycle

The log management process follows a structured lifecycle that ensures logs are properly handled and analyzed. The key stages include:

1. **Collection** – Logs are gathered from various sources such as operating systems, applications, servers, and network devices. Tools like Fluentd and Logstash are commonly used to collect and forward logs to centralized monitoring systems.

2. **Normalization** – Since logs come from different systems with different formats, normalization converts them into a standardized structure such as **JSON or CEF**. This makes it easier to analyze logs and correlate events across multiple sources.

3. **Storage** – After normalization, logs are stored in centralized repositories such as databases or SIEM platforms. Centralized storage allows analysts to access and analyze logs efficiently.

4. **Retention** – Logs are stored for a defined period according to organizational policies or regulatory requirements. Proper log retention ensures that historical data is available for audits, forensic analysis, and compliance.

5. **Analysis** – Security analysts analyze logs to identify unusual patterns, suspicious activities, or security incidents. Analysis is typically performed using SIEM platforms and query languages.

Common Types of Logs

Several types of logs are commonly monitored in cybersecurity environments:

- **Windows Event Logs** – Record system events such as login attempts, account changes, and system errors in Windows operating systems.

- **Syslog** – A standard logging protocol used in Linux and network devices to record system messages and events.

- **HTTP Server Logs** – Generated by web servers such as Apache or Nginx and contain information about web requests, client IP addresses, and server responses.

Practical Implementation

In a laboratory environment, log collection can be implemented using tools like Fluentd or Logstash to gather logs from systems and forward them to a centralized SIEM platform such as the Elastic Stack. Once collected, logs can be normalized into structured formats like JSON and analyzed using query languages.

Security analysts can use **SQL-like query languages**, such as **Kibana Query Language (KQL)** in Microsoft Sentinel or Elastic-based SIEM systems, to search and filter logs for suspicious events. For example, analysts can query authentication logs to detect multiple failed login attempts, which may indicate a brute-force attack.

Through effective log management, organizations gain better visibility into system activities and improve their ability to detect and respond to cybersecurity threats.

I. Log Collection and Normalization Using Fluentd and Elastic SIEM

1. Objective

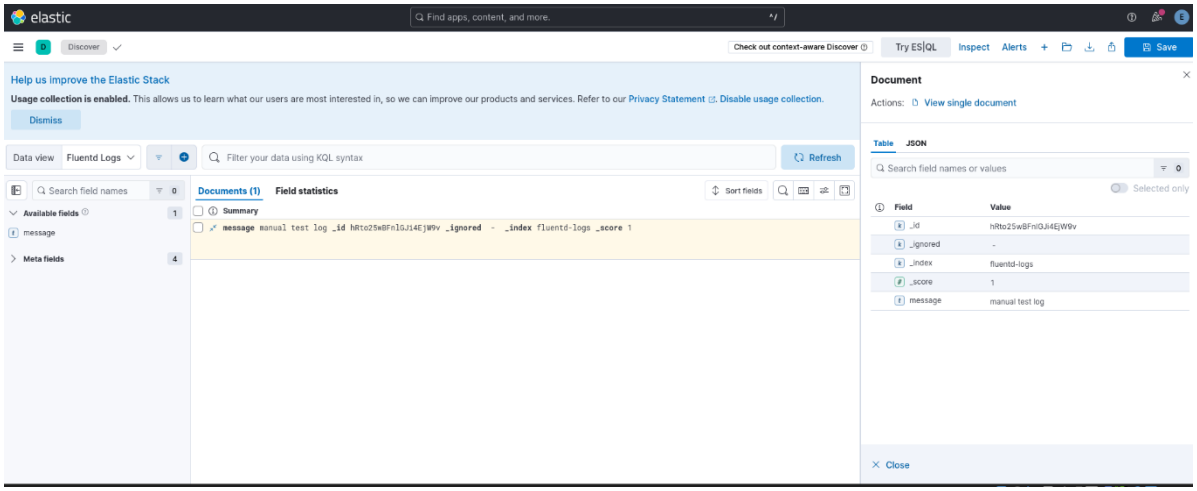
The objective of this exercise is to design and implement a log collection pipeline in a lab environment. The lab demonstrates how system logs can be collected, normalized, and analyzed using a SIEM platform.

The exercise involves configuring a log collector to gather syslog events, forwarding them to a centralized logging platform, and performing queries to analyze security events
.

2. Tools Used

Tool	Purpose
Fluentd	Log collection and forwarding
Elasticsearch	Log storage and indexing
Kibana	Log visualization and analysis
Linux Logger	Generate test logs

3.Pratical Output



II. Snort Rule Testing

To detect HTTP requests to a suspicious domain using Snort custom rules.
The lab includes:

- Network intrusion detection using Snort
- Vulnerability scanning using Nessus
- Endpoint monitoring using Osquery

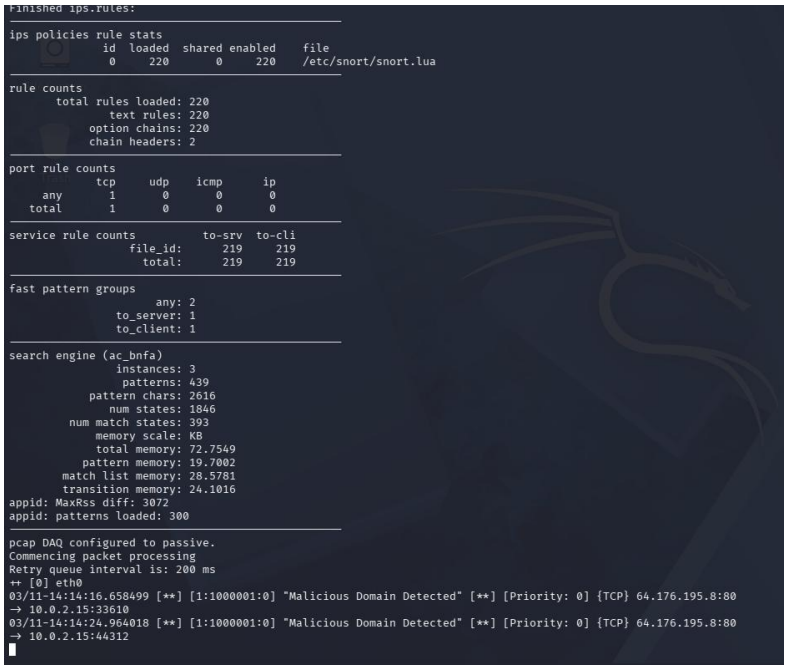
These tools help security analysts detect malicious activity and identify vulnerabilities in systems.

Component	Description
Attacker Machine	Kali Linux
Target Machine	Metasploitable2
Monitoring Tools	Snort
Vulnerability Scanner	Nessus Essentials
Endpoint Monitoring	Osquery
Network Mode	Bridged / NAT

Snort Rule Created

alert tcp any any -> any 80 (msg:"Malicious Domain"; content:"malicious.com"; http_uri; sid:1000001;)

Snort Output

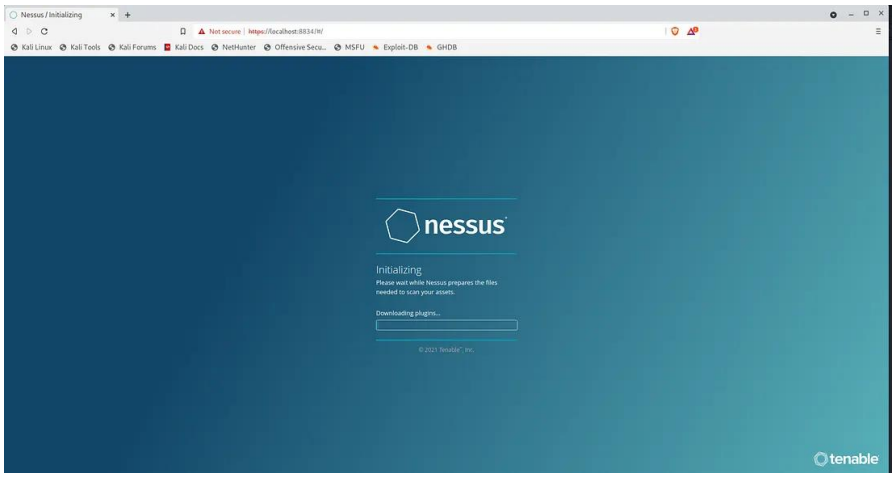


```
Finished ips.rules:
ips policies rule stats
              id  loaded  shared enabled    file
               0     220       0     220    /etc/snort/snort.lua
rule counts
       total rules loaded: 220
                  text rules: 220
              option chains: 220
              chain headers: 2
port rule counts
                  tcp     udp    icmp      ip
     any            1       0       0       0
   total            1       0       0       0
service rule counts          to-srv  to-cli
                  file_id:      219     219
                    total:      219     219
fast pattern groups
                       any: 2
                 to_server: 1
                 to_client: 1
search engine (ac_bnfa)
                 instances: 3
                  patterns: 439
              pattern chars: 2616
                 num states: 1846
           num match states: 393
              memory scale: KB
              total memory: 72.7549
             pattern memory: 19.7002
          match list memory: 28.5781
          transition memory: 24.1016
appid: MaxRss diff: 3072
appid: patterns loaded: 300
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
03/11-14:14:16.658499 [**] [1:1000001:0] "Malicious Domain Detected" [**] [Priority: 0] {TCP} 64.176.195.8:80
-> 10.0.2.15:33610
03/11-14:14:24.964018 [**] [1:1000001:0] "Malicious Domain Detected" [**] [Priority: 0] {TCP} 64.176.195.8:80
-> 10.0.2.15:44312
```

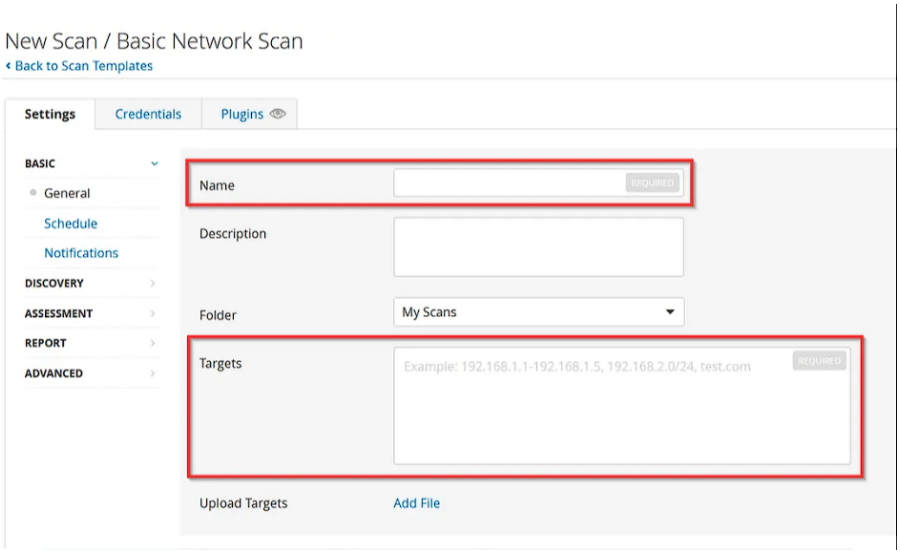
III. Nessus Vulnerability Scan

To identify vulnerabilities in a target system using Nessus Essentials.

Scan Target Machine: **Metasploitable2**



Downloading and Compiling Plugins



Nessus Basic Scan Interface

Nessus Scan output (report and list the top 3 vulnerabilities by CVSS score)

Sev	CVSS	VPR	EPSS	Name	Family	Count
CRITICAL	10.0 *	7.4	0.8622	UnrealIRCd Backdoor Detection	Backdoors	1
CRITICAL	10.0			Canonical Ubuntu Linux SEoL (8.04.x)	General	1
CRITICAL	10.0 *			VNC Server 'password' Password	Gain a shell remotely	1

IV. Osquery Endpoint Monitoring

To monitor system activity and detect potentially malicious processes using Osquery**.**

Installation
Osquery was installed on the Windows virtual machine.

Query Used:

SELECT pid, name, path, cmdline FROM processes WHERE name='cmd.exe' ORDER BY start_time DESC LIMIT 5:

```
C:\Users\Administrator>osqueryi
Using a [1mvirtual database[0m. Need help, type '.help'
osquery> SELECT pid, name, path, cmdline
    ...> FROM processes
    ...> WHERE name='cmd.exe'
    ...> ORDER BY start_time DESC
    ...> LIMIT 5;
+------+---------+---------------------------+-------------------------------+
| pid  | name    | path                      | cmdline                       |
+------+---------+---------------------------+-------------------------------+
| 3964 | cmd.exe | C:\Windows\System32\cmd.exe | cmd  /k malicious_sim_long.bat |
| 5912 | cmd.exe | C:\Windows\System32\cmd.exe | "C:\Windows\system32\cmd.exe"  |
| 6900 | cmd.exe | C:\Windows\System32\cmd.exe | "C:\Windows\system32\cmd.exe"  |
+------+---------+---------------------------+-------------------------------+
osquery>
```

Analysis and Findings

- **Snort successfully detected suspicious HTTP traffic using a custom rule.**
- **Nessus identified multiple critical vulnerabilities in Metasploitable2.**
- **Osquery provided visibility into running processes and detected simulated malicious activity.**

Basic Security Concepts

1.1 CIA Triad
The CIA Triad is the foundation of information security.

Principle	Description	Example
Confidentiality	Ensures only authorized users can access information	Encryption, access control
Integrity	Ensures data is not altered or tampered with	Hashing, digital signatures
Availability	Ensures systems and data are accessible when needed	Redundant systems, backups

1.2 Threat, Vulnerability, and Risk

Term	Definition	Example
Threat	A potential danger that can exploit a system	Malware, hackers
Vulnerability	Weakness in a system	Unpatched software
Risk	Probability that a threat will exploit a vulnerability	Data breach due to unpatched server

IV. Brute-Force Detection Using Windows Logs

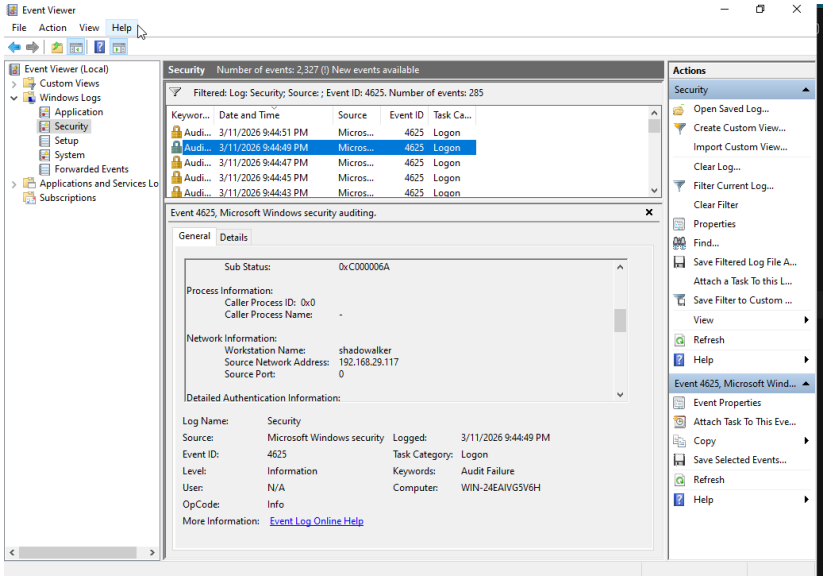
The objective of this task was to simulate multiple failed login attempts and analyze Windows security logs to detect potential brute-force attacks.
Analysis was performed using Windows Event Viewer and the command-line tool wevtutil.

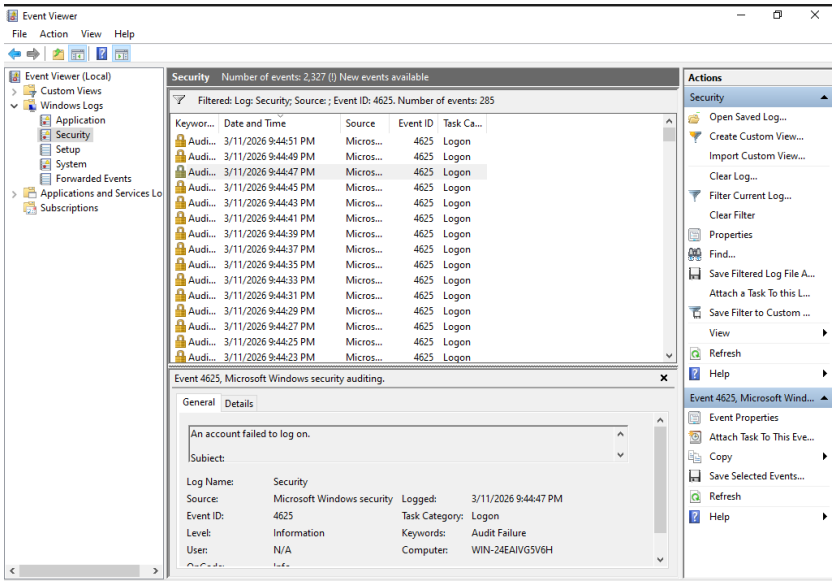
```
┌──(shadowalker㉿shadowalker)-[~]
└─$  hydra -l testuser -P /usr/share/wordlists/rockyou.txt -t 1 rdp://192.168.29.39

Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizati
ons, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-12 10:11:16
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 1 task per 1 server, overall 1 task, 14344399 login tries (l:1/p:14344399), ~14344399 tries per task
[DATA] attacking rdp://192.168.29.39:3389/
[STATUS] 31.00 tries/min, 31 tries in 00:01h, 14344368 to do in 7712:02h, 1 active
[STATUS] 29.67 tries/min, 89 tries in 00:03h, 14344310 to do in 8058:37h, 1 active
[STATUS] 29.71 tries/min, 208 tries in 00:07h, 14344191 to do in 8045:38h, 1 active
[STATUS] 29.53 tries/min, 443 tries in 00:15h, 14343956 to do in 8094:47h, 1 active
[STATUS] 29.52 tries/min, 915 tries in 00:31h, 14343484 to do in 8099:15h, 1 active
[STATUS] 29.51 tries/min, 1387 tries in 00:47h, 14343012 to do in 8100:29h, 1 active
[STATUS] 29.51 tries/min, 1859 tries in 01:03h, 14342540 to do in 8100:58h, 1 active
[STATUS] 29.49 tries/min, 2330 tries in 01:19h, 14342069 to do in 8104:37h, 1 active
[STATUS] 29.49 tries/min, 2802 tries in 01:35h, 14341597 to do in 8104:03h, 1 active
[STATUS] 29.50 tries/min, 3274 tries in 01:51h, 14341125 to do in 8103:35h, 1 active
[STATUS] 29.50 tries/min, 3747 tries in 02:07h, 14340652 to do in 8100:59h, 1 active
[STATUS] 29.50 tries/min, 4218 tries in 02:23h, 14340181 to do in 8102:46h, 1 active
[STATUS] 29.49 tries/min, 4689 tries in 02:39h, 14339710 to do in 8104:08h, 1 active
[STATUS] 29.49 tries/min, 5161 tries in 02:55h, 14339238 to do in 8103:38h, 1 active
[STATUS] 29.49 tries/min, 5633 tries in 03:11h, 14338766 to do in 8103:10h, 1 active
[STATUS] 29.49 tries/min, 6105 tries in 03:27h, 14338294 to do in 8102:44h, 1 active
```

a. Checking the Brute force Attack via Event Viewer





The Above Attachments Shows that the 4625 event where the source information is visible

b. Event Viewer Filtered (4625 event only) exerted to csv data

	A	B	C	D	E	F
1	Keywords	Date and Time	Source	Event ID	Task Category	
2	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
3	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
4	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
5	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
6	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
7	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
8	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
9	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
10	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
11	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
12	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
13	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
14	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
15	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
16	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
17	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
18	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
19	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
20	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
21	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
22	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
23	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
24	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
25	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.
26	Audit Failure	03-11-2026 21:44	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on.

c. Checking the Brute force Attack via wevtutil CLI.

Select Administrator: Command Prompt

```
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2026-03-12T01:27:15.5500000Z
  Event ID: 4625
  Task: N/A
  Level: N/A
  Opcode: N/A
  Keyword: N/A
  User: N/A
  User Name: N/A
  Computer: WIN-24EAIVG5V6H
  Description:
N/A

Event[6847]
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2026-03-12T01:27:17.5750000Z
  Event ID: 4625
  Task: N/A
  Level: N/A
  Opcode: N/A
  Keyword: N/A
  User: N/A
  User Name: N/A
  Computer: WIN-24EAIVG5V6H
  Description:
N/A
```

Same above data in a test format

```
Event[0]
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2026-03-04T06:32:22.2300000Z
  Event ID: 4625
  Task: Logon ?  Level: Information ?  Opcode: Info  ?  Keyword: Audit Failure  ?  User: N/A
  User Name: N/A
  Computer: WIN-24EA2VGSV6H
  Description:
An account failed to log on.
Subject:
        Security ID:            S-1-5-21-2619178725-3750698332-2456648091-500
        Account Name:           Administrator
        Account Domain:         WIN-24EA2VGSV6H
        Logon ID:               0xF3E8F

Logon Type:                     2

Account For Which Logon Failed:
        Security ID:            S-1-0-0
        Account Name:           Administrator
        Account Domain:         WIN-24EA2VGSV6H

Failure Information:
        Failure Reason:         Unknown user name or bad password.
        Status:                 0xC000006D
        Sub Status:             0xC000006A

Process Information:
        Caller Process ID:      0x4E8
        Caller Process Name:    C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe

Network Information:
        Workstation Name:       WIN-24EA2VGSV6H
        Source Network Address: -
        Source Port:            -

Detailed Authentication Information:
        Logon Process:          Advapi
        Authentication Package: Negotiate
        Transited Services:     -
        Package Name (NTLM only):       -
        Key Length:             0

This event is generated when a logon request fails. It is generated on the computer where access was attempted.

The Subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The Logon Type field indicates the kind of logon that was requested. The most common types are 2 (interactive) and 3 (network).

The Process Information fields indicate which account and process on the system requested the logon.

The Network Information fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.

The authentication information fields provide detailed information about this specific logon request.
        - Transited services indicate which intermediate services have participated in this logon request.
        - Package name indicates which sub-protocol was used among the NTLM protocols.
        - Key length indicates the length of the generated session key. This will be 0 if no session key was requested. ?
```

Test Procedure

1. **Multiple incorrect login attempts were intentionally performed on the Windows virtual machine.**
2. **Each failed login attempt generated a Windows Security Event ID 4625.**
3. **Security logs were opened in Event Viewer and filtered using Event ID 4625.**
4. **The filtered results were exported to a CSV file for analysis.**

d. Security Event Documentation

*1.**Incident Documentation Table***

Date/Time	Source IP	Event ID	Description	Action Taken
03-11-2026 21:44	192.168.29.x	4625	Multiple failed login attempts detected. Event recorded as “An account failed to log on.” indicating a possible brute-force attack.	Security logs reviewed and suspicious login activity documented. Monitoring enabled for further failed login attempts.
03-11-2026 21:44	192.168.29.x	4625	Multiple failed login attempts detected. Event recorded as “An account failed to log on.” indicating a possible brute-force attack.	Security logs reviewed and suspicious login activity documented. Monitoring enabled for further failed login attempts.
03-11-2026 21:43	192.168.29.x	4625	Multiple failed login attempts detected. Event recorded as “An account failed to log on.” indicating a possible brute-force attack.	Security logs reviewed and suspicious login activity documented. Monitoring enabled for further failed login attempts.
03-11-2026 21:43	192.168.29.x	4625	Multiple failed login attempts detected. Event recorded as “An account failed to log on.” indicating a possible brute-force attack.	Security logs reviewed and suspicious login activity documented. Monitoring enabled for further failed login attempts.

*2. **Security Event Explanation***

Event Source: Logs were collected from Windows Event Viewer under the Security log.

Event Details:

- Event ID: 4625
- Task Category: Logon
- Keyword: Audit Failure
- Message: “An account failed to log on.”

V. SOC Monitoring and Alerting Implementation Report

1 . Objective

The objective of this task was to implement monitoring dashboards and alert rules in a Security Information and Event Management (SIEM) environment. The lab simulated security incidents such as brute-force login attempts and analyzed the generated logs to detect suspicious activities.
The monitoring infrastructure was implemented using Elastic SIEM, Wazuh, and Osquery to emulate a mini Security Operations Center (SOC) environment.

2. Tools and Technologies Used

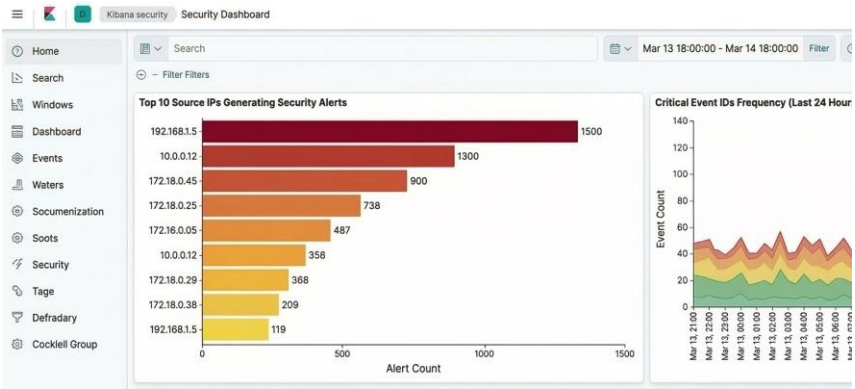
Tool	Purpose
Elastic SIEM	Log analysis and security monitoring
Kibana	Visualization dashboards
Wazuh	Host-based intrusion detection
Osquery	Endpoint visibility and telemetry
Kali Linux	Attack simulation environment
Windows VM / Linux VM	Target systems for monitoring
SSH	Simulating brute-force login attempts

3. Monitoring Dashboard Configuration

Monitoring dashboards were created to visualize suspicious activities and security events in the SIEM environment.

A . Dashboard 1: Top Source IPs Generating Alerts

This dashboard identifies IP addresses that generate the highest number of security alerts.



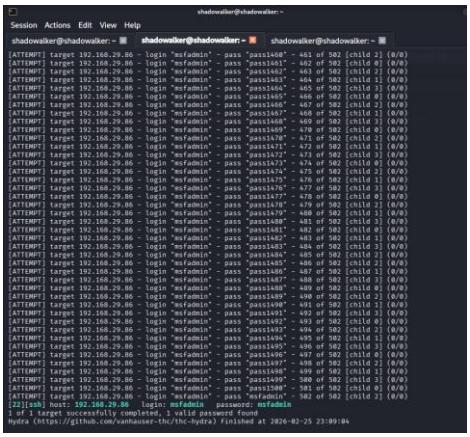
B. Alert Rule Configuration in Elastic SIEM

An alert rule was configured to detect brute-force login attempts.

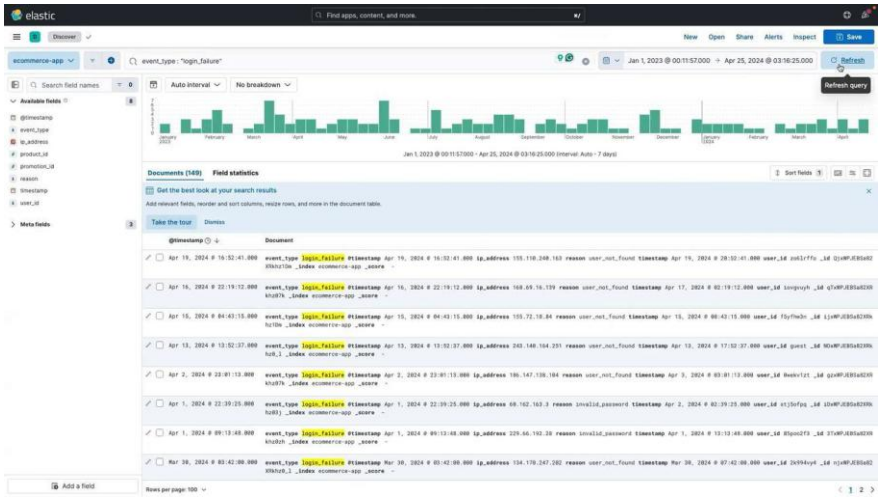
Rule Name
Detect Multiple Failed Logins

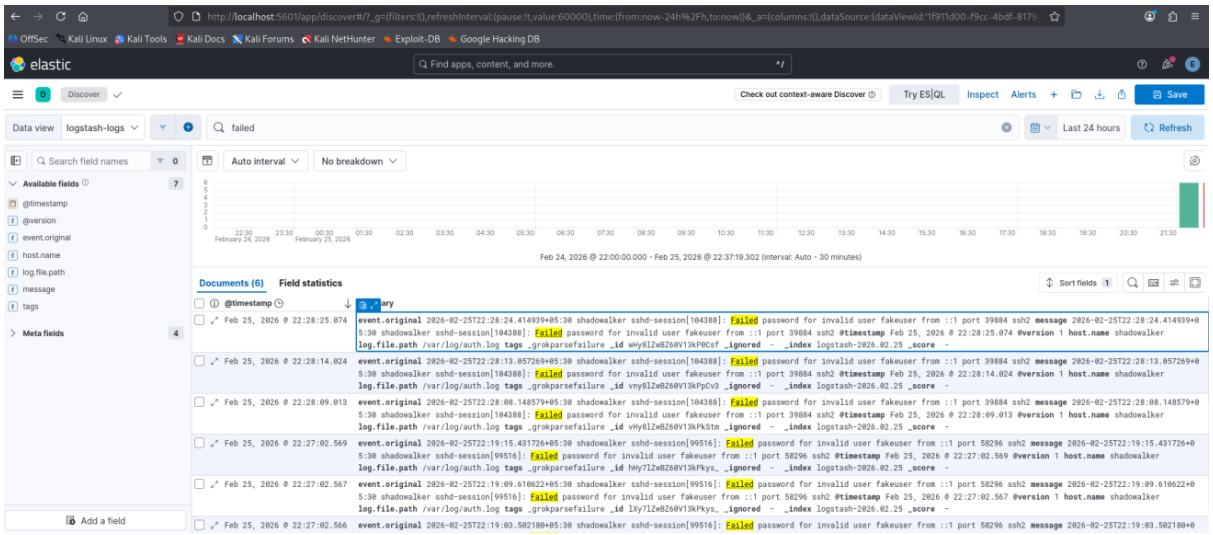
Configuration

Parameter	Value
Index	security-login-*
Query	event.code:4625
Threshold	Count > 5
Time Window	5 minutes



Doing a Brute Force Attack



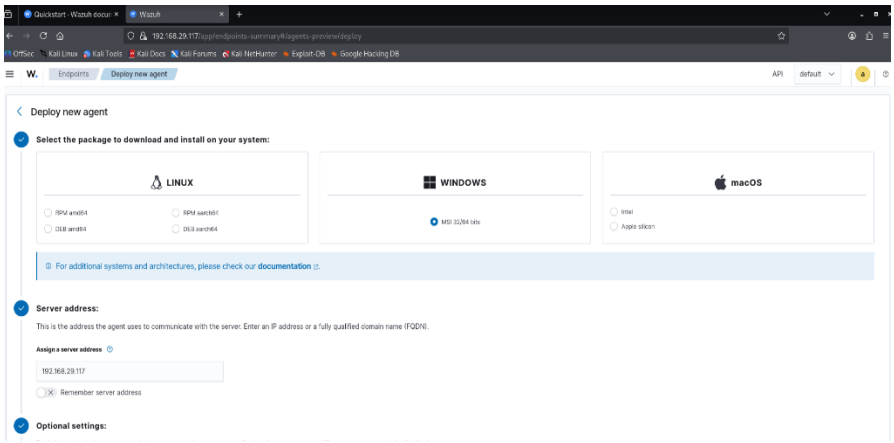


Advanced Detection Rule using Wazuh

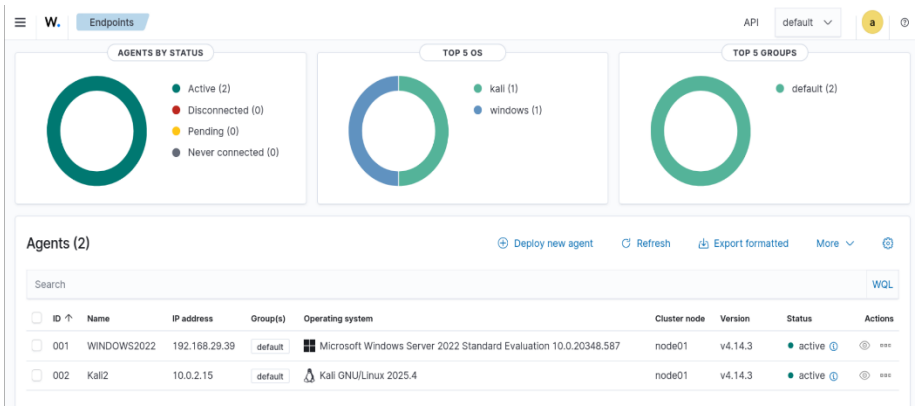
A custom rule was implemented in Wazuh to detect SSH brute-force attempts.
Rule Logic

Trigger alert when:

- **3 or more failed login attempts occur within 2 minutes**



Deploy a Wazuh agent on windows



Wazuh after deploying agents

.

Testing Procedure

The rule was tested by repeatedly attempting SSH logins using incorrect passwords.

Command used:

ssh user@192.168.1.x

Multiple failed attempts triggered the Wazuh alert rule.

Alert Monitoring in Wazuh Dashboard

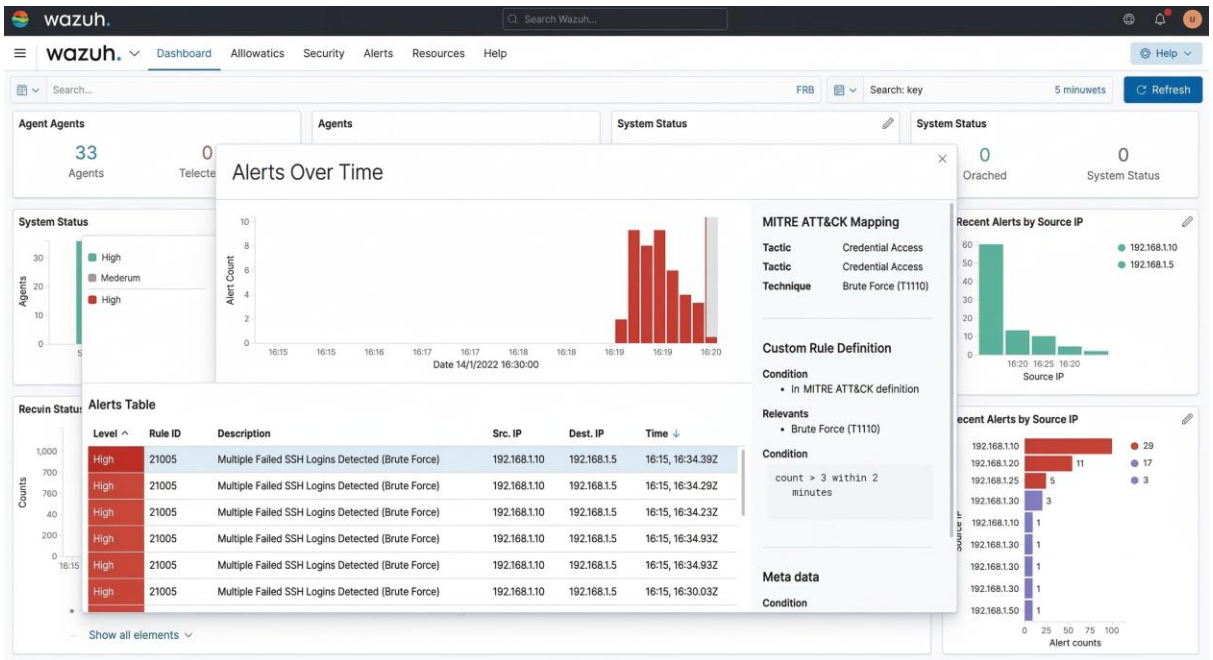
After the attack simulation:

1. Wazuh collected authentication logs.
2. Failed login attempts were detected.
3. Alerts were displayed in the Wazuh dashboard.

The dashboard displayed:

- Alert severity
- Source IP
- Username targeted
- Timestamp of the event

This confirmed that the custom detection rule successfully identified brute-force login attempts.



VI . MITRE ATT&CK Mapping

The detected security events in the monitoring dashboard were mapped to the MITRE ATT&CK framework to classify attacker behavior and identify the tactics used during the simulated attack scenarios.

The dashboard uses pre-built Sigma detection rules integrated with the SIEM platform to automatically correlate alerts with relevant MITRE ATT&CK techniques. This mapping helps SOC analysts understand how attackers progress through different stages of an attack lifecycle.

Observed Techniques from the Dashboard

Technique	MITRE ID	Description
Valid Accounts	T1078	Attackers attempt to gain access using legitimate user credentials. This can occur during brute-force attempts or credential reuse attacks.
Command and Scripting Interpreter	T1059	Attackers execute commands through scripting environments such as PowerShell, Bash, or SSH shells to control compromised systems.
Successful Logon Activity	T1024	Indicates successful authentication events that may represent legitimate access or compromised account usage.
Failed Logon Attempts	T4625	Multiple failed authentication attempts indicating possible brute-force or password spraying attacks.
Privilege Escalation	T1089	Attempts to gain higher system privileges after initial access to expand attacker capabilities.
Credential Access	T1082 / T1078	Techniques used to obtain or reuse credentials to maintain persistence or move laterally within a network.

Observations from the Security Dashboard

The dashboard visualization shows:

- **Top Source IPs Generating Security Alerts**, indicating hosts producing the highest number of suspicious events.
- **Critical Event ID Frequency**, highlighting spikes in authentication-related activities such as successful and failed logins.
- **MITRE ATT&CK Technique Mapping**, which categorizes alerts according to attacker tactics including reconnaissance, initial access, execution, persistence, privilege escalation, and credential access.

For example, the dashboard detected:

- Repeated **failed login attempts**, which align with the **Brute Force (T1110)** technique.
- Execution-related activity mapped to **Command and Scripting Interpreter (T1059)**.
- Alerts related to **Valid Accounts (T1078)** indicating possible credential misuse.

✓ **Importance of MITRE ATT&CK Mapping**

Mapping alerts to the MITRE ATT&CK framework provides several benefits:

1. Helps SOC analysts understand attacker behavior in a standardized format.
2. Enables correlation between different stages of an attack lifecycle.
3. Improves threat detection and response capabilities.
4. Allows security teams to prioritize incidents based on adversary tactics.

By integrating MITRE ATT&CK mapping within the SIEM dashboards, analysts can quickly identify potential attack patterns and investigate suspicious activities more effectively.

Conclusion

This project provided a comprehensive understanding of Security Operations Center (SOC) fundamentals and practical security monitoring techniques. The study combined both theoretical knowledge and hands-on practice to simulate real-world SOC operations. Through the exploration of SOC roles, workflows, and monitoring objectives, the project established a strong foundation in how modern security teams detect, analyze, and respond to cyber threats.

From a theoretical perspective, key concepts such as the SOC operational structure, security monitoring principles, log management processes, and incident response lifecycle were studied. Understanding frameworks like the NIST Incident Response Framework and the MITRE ATT&CK framework helped in identifying attacker behaviors and mapping them to standardized threat techniques. Additionally, important security concepts such as the CIA triad, defense-in-depth, and zero trust architecture provided insight into how organizations design secure environments.

The practical component of the project focused on implementing a mini SOC laboratory environment using open-source tools. Platforms such as Elastic SIEM, Wazuh, and Osquery were used to collect, analyze, and monitor security logs. Various tasks were performed including log collection, log analysis, event monitoring, and alert configuration. By analyzing

Windows security logs, particularly events such as failed login attempts (Event ID 4625), the project demonstrated how suspicious activities like brute-force attacks can be detected in real time.

Monitoring dashboards were created to visualize critical security data such as the top source IP addresses generating alerts and the frequency of important security events. These dashboards provide SOC analysts with situational awareness and help prioritize investigations more efficiently.

Alert rules were also implemented to automate threat detection. For example, rules were configured to trigger alerts when multiple failed login attempts occur within a defined time window. These rules were tested through simulated attacks such as repeated SSH login attempts with incorrect credentials. The alerts generated in the SIEM platform confirmed that the monitoring and detection mechanisms were functioning correctly.

Furthermore, security events were documented using standardized reporting templates to simulate SOC documentation practices. Proper documentation ensures that incidents are recorded accurately and that security teams can review and improve their response strategies.

Overall, this project successfully demonstrated the workflow of a SOC environment, including log collection, threat detection, alerting, investigation, and documentation. The integration of theoretical knowledge with practical implementation helped develop a deeper understanding of how SOC analysts monitor systems, detect threats, and respond to security incidents. The experience gained from this project serves as a strong foundation for further learning in cybersecurity operations, threat hunting, and incident response.